Development of an Adaptive Model to Assess Suitability of Hyperloop Systems Across Diverse Loci

Undergraduate Research Dissertation (2026 European Hyperloop Week: Groningen)



SMITH ENGINEERING
HYPERLOOP
Queen's University
Design Team

Authors:

Saksham Mirchandani, Lucas Horacsek, Pranav Tripathi, Shaurya Dugala

Contributors:

Matthew Stevenson, Mel Choloniuk, Dr. Ryan Grant

Submission Date: 14 June 2026

Word Count: 5122

Award: Full-Scale Award – Socio-Economic Aspects of Hyperloop Systems

Overview

The Smith Hyperloop Design Team consists of 120 undergraduate students, divided among four different departments, each with specialized teams. One of these departments is Research and Development, which oversees the Business Competition research team. The Business Competition team consists of a Project Director, Project Manager, and four Research Specialists. The team is located at the Integrated Learning Centre (ILC) on Queen's University campus.

Statement of Originality

As a part of following professional engineering practices in Canada, we bear the burden of proof for original work. We confirm that the work within this report is original, created by various members on our team, and sources are cited appropriately where required.

List of Team Members

The research submitted to the Non-Technical Research Paper Award is conducted by the Business Competition Research Team of the Research and Development department of the Smith Hyperloop Design Team. The team members and contributors of this submission are listed below:

Name	Position	Contact
Saksham Mirchandam	Project Manager	Mirchandani.saksham@queensu.ca
Lucas Horacsek	Research Specialist	22mbs6@queensu.ca
Pranav Tipathi	Research Specialist	Pranav.tripathi@queensu.ca
Shaurya Dugala	Research Specialist	22qp48@queensu.ca
Matthew Stevenson	Director of Research and Development	20mwss@queensu.ca
Melissa Choloniuk	Team Co-Captain	15mmc3@queensu.ca

Development Environment and Research Objectives

The objective of the Research and Development department is to explore innovations for integration into Hyperloop systems. To accomplish this, it conducts substantial research alongside partners from other institutions for knowledge while corroborating the validity of the research.

Design Competition Award

This research paper is intended for the Design Award: *Full-Scale Award – Socio-Economic Research.*

Abstract

Advancements are fast being made to Hyperloop technology due to the continual urge to improve and optimise, while adequate holistic research lags. Current studies focus on engineering feasibility with more abstract concepts such as corridor selection and configuration planning have received far less rigorous inquiry. The solution proposed herein is contingent on a framework to automatically evaluate corridor selection and passenger-freight allocation. The model is designed with versatility in mind, able to be superimposed on any number of potential routes, assuming sufficient data is available. The framework is validated with a collection of case studies, wherein the model is applied to scenarios covering a wide breadth of edge cases: cross-border routes with intermediate nodes, high-density domestic corridors with existing modal demand, and multi-national routes with established regulatory and investment frameworks.

The methodology draws on two types of input data. The first is geographical data: terrain profile, corridor distance, elevation change, and right-of-way constraints. The second is socio-economic data: city-level population, gross domestic products, existing travel demand by mode, freight industry composition, and cross-border regulatory environment. Together, these inputs are used to assess and compare corridors, providing a basis for identifying optimal routes across a given region and illustrating the value of a Hyperloop system therein.

Hyperloop infrastructure supports both passenger and freight vehicles on the same guideway. The freight focus is on time-sensitive cargo (e.g. just-in-time supply chains, pharmaceuticals, and high-value manufacturing), wherein faster transit times translate directly into cost savings for businesses along the corridor. Findings indicate that corridor viability is most sensitive to economic connectivity density and freight integration potential, rather than to distance or capital cost alone. Corridors with proximity to major logistic nodes, such as large ports or established distribution hubs, benefit from stable freight demand that improves utilisation across the full network.

Table of Contents

1. Introduction *1*

2. Literature Review *2*

2.1 Socio-Economic Context of Hyperloop Corridor Development **2**

2.1.1 Economic Feasibility and Cost Structures 2

2.1.2 Cross-Border Governance and Regulatory Frameworks 3

2.1.3 Migration, Regional Development, and Equity 4

2.2 Dijkstra's Algorithm and Its Application to Network Routing **5**

2.2.1 Graph-Theoretic Foundations 5

2.2.2 Dijkstra's Algorithm 6

2.2.3 Relevance to Hyperloop Corridor Optimization 7

3. Methodology *7*

3.1. Technical Implementation Overview **7**

4. Results *9*

4.1. West Africa: Lagos-Accra Corridor (~550 km) **9**

4.1.1. Socio Economic Context 9

4.1.2. Transportation Landscape 10

4.1.3. Regulatory and Policy Context 10

4.1.4. Benefits of Implementing Hyperloop 11

4.2. Southeast Asia: Jakarta-Surabaya Corridor (~740 km) **11**

4.2.1. Socio Economic Context 11

4.2.2. Transportation Landscape 11

4.2.3. Regulatory and Policy Context 12

4.2.4. Benefits of Implementing Hyperloop 12

4.3. The Middle East: The Gulf **12**

4.3.1. Socio Economic Context 12

4.3.2. Transportation Landscape 13

5. Conclusion and Next Steps *13*

References *16*

Appendix 1: Node Scoring Weight Profile *20*

Appendix 2: Lagos-Accra corridor geographic region of interest *21*

Appendix 3: Lagos-Accra Corridor: Node Clusters Output *21*

Appendix 4: Lagos–Accra Corridor - Optimal Hyperloop Node Network *22*

1. Introduction

The global transportation sector stands at an inflection point. Rising urban populations, mounting pressure on existing aviation and rail networks, and the accelerating demand for time-sensitive freight movement have together exposed the limitations of conventional high-speed transport. Hyperloop technology, which propels passenger and cargo vehicles through a low-pressure tube at speeds exceeding 700 km/h, has emerged as a candidate solution to these systemic constraints. Yet the dominant body of research on hyperloop remains concentrated on engineering feasibility, covering tube pressure dynamics, propulsion architecture, and structural alignment, while the prior question of which corridors are most viable, and in what passenger-freight configuration, has received comparatively limited rigorous treatment [1].

This gap is consequential. Infrastructure investment decisions of this scale carry long-term economic and political commitments that are difficult to reverse. A corridor selected on engineering grounds alone, without accounting for existing travel demand, freight industry composition, economic connectivity between endpoints, and cross-border regulatory conditions, risks misallocating capital and underperforming on utilization. Hyperloop systems depend on high and consistent throughput to justify their construction costs, which early estimates have placed in the range of 17 to 27 million USD per mile, figures comparable to or exceeding those of conventional high-speed rail on equivalent routes [2]. The corridor selection problem is therefore not an ancillary design question but a fundamental determinant of the value generation of a Hyperloop system.

This paper addresses that gap through a corridor selection and passenger-freight allocation framework grounded in quantifiable socio-economic and geographic data. The framework draws on two categories of input. The first encompasses objective geographic data, including terrain profile, corridor distance, elevation change, and right-of-way constraints. The second encompasses socio-economic data, including city-level population, gross domestic product, existing travel demand by mode, freight industry composition, and cross-border regulatory environment. Together, these inputs are used to compare corridors and identify the configurations under which a Hyperloop system generates the greatest network value [3].

The framework is applied to three corridors selected to represent distinct structural archetypes. The Toronto-Montreal corridor embodies a high-density domestic route with strong existing modal demand, natural intermediate nodes, and an uncomplicated regulatory environment. The London-Paris corridor represents a cross-border route embedded in an active infrastructure framework, where post-Brexit border policy introduces additional legal complexity. The Riyadh-Dubai corridor represents a high-growth, business-travel-oriented route embedded in a regional policy environment increasingly oriented toward economic diversification and frictionless cross-border mobility. The balance of this report is organized as follows. The literature review establishes the socio-economic and algorithmic foundations of the framework. The methodology section details the data inputs, corridor assessment criteria, and allocation model. The results present the comparative corridor analysis and allocation outputs. The discussion interprets those findings relative to existing literature and evaluates the broader applicability of the framework.

2. Literature Review

2.1 Socio-Economic Context of Hyperloop Corridor Development

2.1.1 Economic Feasibility and Cost Structures

Cost-benefit analyses of proposed Hyperloop projects consistently reveal a tension between substantial long-term economic gains and significant upfront capital requirements. A feasibility study of the Chicago-Columbus-Pittsburgh corridor projected 300 billion USD in economic benefits over 30 years, driven by travel time savings, reduced private vehicle use, and lower emissions, alongside 1.9 billion diverted car trips and meaningful reductions in CO_2 output [4]. The projected gains must be weighed against multibillion-dollar construction costs, and robust public-private financing structures pose a viable means by which to realize such projects.

Construction cost estimates for Hyperloop infrastructure have ranged between 17 and 27 million USD per mile, excluding land acquisition, placing individual corridors in a cost class comparable to, and in some cases exceeding, conventional high-speed rail on equivalent routes [5]. Transport Canada's preliminary feasibility assessment reached a similar conclusion, indicating that Hyperloop user costs would resemble airfares rather than rail fares, with passenger pricing

estimated around 0.20 USD per mile, or roughly 60 USD for a 300-mile trip [6]. Modelling commissioned for the Midwest corridor suggested an annual ridership base exceeding two million passengers if fare targets are achieved, indicating that travellers are generally willing to pay aviation-equivalent prices for substantially faster ground transport [4].

Infrastructure cost optimization remains an active area of inquiry. Conceptual studies, including a NASA-commissioned Volpe Centre analysis, have identified underwater and underground tube alignments as potentially cost-reducing strategies in specific geographies, as these approaches can avoid expensive land acquisition and, in the case of subaqueous tunnels, leverage external pressure differentials to reduce structural requirements [2]. Shorter route alignments via tunnelling through terrain rather than routing around it may similarly reduce total construction costs, though the technical complexity of such solutions introduces its own risks.

2.1.2 Cross-Border Governance and Regulatory Frameworks

Hyperloop systems designed to cross international borders introduce a category of institutional complexity that has no direct precedent in existing transport governance. Passengers and freight moving at Hyperloop speeds cannot be processed through conventional border inspection regimes without negating the inherent benefits of said speed. The original Hyperloop concept assumed airport-style security protocols, but analysts have noted that a rail-like border control model incorporating pre-clearance and digital checkpoints would be far more compatible with high-frequency Hyperloop operations [2].

The European Union has emerged as the most advanced jurisdiction in developing a formal regulatory framework for Hyperloop. The European Commission has been developing unified health, safety, and environmental standards for Hyperloop operations, with the explicit aim of supporting investment and enabling a continent-wide cross-border network [7]. The Hyperloop Development Programme's position paper framed this regulatory work as a precondition for attracting the private capital necessary to advance commercial deployment [8]. The European framework is widely regarded as a candidate blueprint for global regulatory harmonization, given the precedent-setting role the EU has played in cross-border transport governance for both rail and aviation [3].

Gulf Cooperation Council Unified Visa, which launched in 2026 and covers all six member states under a single tourist permit modelled on the Schengen visa, represents a concrete example of the kind of policy alignment that would benefit Hyperloop corridors crossing between Saudi Arabia, the UAE, and neighbouring states [9]. The initiative displays the importance of cross-border passenger movement and highlights how it is directly complementary to Hyperloop's high-speed form of transportation. Analysts have noted that Hyperloop infrastructure could, in turn, incentivize further modernization of border clearance systems, including expanded e-visa frameworks and biometric pre-clearance, to handle the rapid and high-frequency cross-border flows facilitated by Hyperloop [10].

The emergence of new modes of cross-border transport also raises broader questions concerning requisite international governance. The development of commercial aviation produced the International Civil Aviation Organization, while European railway expansion produced a series of intergovernmental technical and operational agreements. Hyperloop may similarly necessitate bilateral or multilateral accords covering regulatory oversight, emergency response coordination, liability, and revenue-sharing arrangements for jointly operated corridors [3]. The degree to which regional blocs can reach such agreements in advance of commercial deployment will be a significant determinant of how quickly viable cross-border corridors can be activated [7].

2.1.3 Migration, Regional Development, and Equity

High-speed transport infrastructure has a well-documented relationship with urban economic geography. The academic literature on high-speed rail consistently finds that accessibility gains are largest when individual lines are integrated into networks rather than operated as standalone corridors, with Europe and China representing the clearest examples of systems where network maturity produced disproportionate economic returns relative to isolated route performance [11]. The economic and accessibility benefits of Hyperloop are therefore expected to arise not from individual corridors in isolation but from how those corridors interconnect to form a network.

At the municipal and regional level, Hyperloop's capacity to compress effective travel time could reshape commuting patterns and labour market catchment areas in ways that blur the conventional geographic boundary between metropolitan regions. Studies have suggested that

Hyperloop expansion could substantially enlarge 60-minute commuting zones, potentially linking labour markets currently too distant for daily commuting. This enhanced connectivity has the potential to drive urban growth and redistribute economic activity, enabling workers to choose residences further from labour hubs, and even across international boundaries [11]. Whether the effect of such redistribution favours the depopulation of suburban and rural areas through improved accessibility or accelerates the concentration of talent and activity in major hubs that become more reachable, is subject to the corridors in question as well as existing settlement patterns.

Questions of equitable access are central to the political economy of Hyperloop investment. If ticket prices are set at levels comparable to airfare, Hyperloop risks alienating low- to middle-income households, raising legitimate concerns about necessity of public subsidies for infrastructure catered to specific echelons of society [6]. Policymakers considering Hyperloop investment will need to assess subsidy structures, price controls, and the distribution of socio-economic benefits alongside construction cost considerations. Secondary impacts, including land displacement associated with right-of-way acquisition (ROW) and environmental disruption during construction, represent additional equity dimensions that public agencies will need to address [3].

2.2 Dijkstra's Algorithm and Its Application to Network Routing

2.2.1 Graph-Theoretic Foundations

The problem of identifying optimal routes through a transportation network is formalized in graph theory as the shortest path problem. A graph is defined as an ordered pair $G = (V, E)$, where V is a non-empty set of vertices or nodes and E is a set of edges connecting pairs of nodes [12]. In urban and intercity transportation networks, nodes represent intersections, stations, or hubs, while edges represent the routes connecting them, weighted by some measure of traversal cost, typically distance, travel time, or some combination.

Transportation networks introduce two structural properties that have direct implications for routing. The first is directionality: many transportation links operate in only one direction, requiring that the underlying graph be directed, meaning edges carry an orientation from one

node to another [12]. The second network property is weighting: edges are not uniform but carry weights that reflect the real cost of traversal. In urban traffic applications, edge weights have been modelled to incorporate segment length, average traffic flow speed, and delay time at signalized intersections [12]. In the context of Hyperloop, analogous weights would incorporate segment travel time at cruising speed, switching delays at network nodes, and the operational characteristics of freight versus passenger vehicles sharing the same guideway infrastructure.

2.2.2 Dijkstra's Algorithm

Dijkstra's algorithm, first proposed by Edsger W. Dijkstra in 1959, is among the most widely applied methods for solving the single-source shortest path problem on graphs with non-negative edge weights [12]. The algorithm is classified as a greedy method: meaning that at each decision step, the locally optimal choice is selected – from a set of candidate nodes, the algorithm systematically selects the paths with the lowest cumulative cost between each node and the source node [13]. This yields a worst-case time complexity of $O(N^2)$ for a network of N nodes. The logic of the current version of the algorithm begins with an initial distance value of zero is assigned to the source node and infinity to all others. At each iteration, the unvisited node with the smallest known cumulative distance is selected, its neighbours are examined, and their distance values are updated if a shorter path through the current node is identified. This process repeats until the destination node receives a permanent, immutable distance label [12]. The sequence of nodes satisfying this condition from source to destination constitutes the shortest path. Formally, for a node j adjacent to the currently examined node i, the update rule is:

$$d_j = \min\left(d_j, d_i + c_{ij}\right) \tag{1}$$

where d_j is the current best-known distance to node j, d_i is the finalized distance to node i, and c_{ij} is the edge weight between nodes i and j [12], [14]. A key constraint of the algorithm is that it does not support negative edge weights, as negative-cost edges would violate the greedy assumption that a finalized node cannot be improved by future exploration. In transportation applications, this constraint is not binding, as measurable costs such as distance, time, or energy consumption are always non-negative values.

2.2.3 Relevance to Hyperloop Corridor Optimization

Dijkstra's algorithm provides a principled basis for identifying the minimum-cost path between any origin-destination pair, considering the full set of operational constraints that govern transport. A Hyperloop network can be represented as a weighted directed graph wherein nodes constitute stations and hubs, edges represent guideway segments between them, and edge weights reflect the generalized cost of traversal, incorporating travel time, switching delays, and operational constraints specific to passenger or freight vehicle types [12]. Grujic and Grujic (2025) demonstrate this approach in an urban traffic context, applying the algorithm to a network with one-way streets, average flow speeds, and intersection delay times as composite edge weights, and validating results against Google Maps routing data for Sarajevo, Bosnia and Herzegovina. The methodological analogy to Hyperloop is direct: high-speed rail networks similarly involve guided movement, variable traversal costs, and hub delay considerations that translate intuitively to a weighted graph framework.

Beyond single-corridor path optimization, Dijkstra's algorithm is relevant to network-level routing problems involving multiple origin-destination pairs and mixed vehicle types. In Hyperloop applications, wherein passenger and freight vehicles share mainline infrastructure but differ in loading time, time sensitivity, and value density, routing becomes an issue of allocating capacity across competing flows while minimizing total network traversal cost [11], [12]. This formulation, in which node-level allocation decisions determine both the regions benefiting from the network and how freight and passenger flows interact, forms the analytical basis of the methodology described herein.

3. Methodology

3.1. Technical Implementation Overview

The corridor selection tool is implemented as a modular Python pipeline, wherein each stage operates on a shared list of node objects, each carrying an ID, name, coordinates, and a dictionary of attributes that grows as the node passes through later stages. The pipeline is

orchestrated from *main.py*, which calls each stage in sequence, taking the system from raw geographic data through to a selected set of stations and the resulting network graph.

The pipeline begins with candidate discovery. Using the OSMnx library, the system queries OpenStreetMap for all populated places (tagged as cities, towns, villages, or hamlets) that fall within a bounding box defined in the configuration file. For each result returned, the centroid of its geometry is taken as the node coordinates, and its name and population are extracted where available.

Each candidate node then passes through an enrichment stage, which attaches the four attributes used for scoring. Elevation variance is computed by querying the Open-Elevation API at the node's location plus four offset points roughly 2 km away in each cardinal direction and taking the range between the highest and lowest elevation returned as a proxy for terrain ruggedness; flatter terrain scores better, as it is cheaper and easier to build through. Transport connectivity is calculated by querying OpenStreetMap for airports and rail stations (stations and halts) within the bounding box, then counting how many of these fall within 10 km of each node using geodesic distance; a high count indicates a location that is already well-served by other transport modes, while a low count indicates an underserved location where Hyperloop adds more marginal value. Port proximity queries OpenStreetMap for harbours and port-zoned land within the bounding box and computes the geodesic distance from each node to the nearest one, reflecting freight integration potential. Distance to the nearest major city is computed without any external API call: nodes whose population exceeds a configurable threshold are treated as "major cities," and every node's distance to the nearest qualifying city (excluding itself) is calculated directly from the node list. This acts as a proxy for passenger demand.

Once enrichment is complete, each of the four attributes is normalized independently across all nodes using min-max scaling and mapping each onto a 0–1 range. This ensures sufficient comparison and combination. A weighted score is then computed for each node by summing across the four attributes. The normalized value is triple that attribute's weight from the active weight profile. For attributes where a lower raw value is better (elevation variance, distance to port, and distance to major cities), the normalized value is inverted (1

- norm) before being weighted. The result is that every case providing a higher contribution to the score reflects a more desirable outcome.

With every node scored, K-means clustering is applied to the coordinates of all candidate nodes, dividing them into n clusters, wherein n is the target number of stations. Each node is labelled with its cluster, and from each cluster the single highest-scoring node is selected, producing a final shortlist of stations that are both high-scoring and spread across the bounding box rather than concentrated in one area. The full candidate set and the final selected set can both be plotted, coloured and labelled by cluster, and saved as image files. The selected stations are assembled into a NetworkX graph where an edge is added between any two stations. The system computes the shortest path between any two stations in this graph using A* search. The geodesic distance to the target station is used as the search heuristic. The resulting network, representing stations as nodes and feasible direct connections as weighted edges, is plotted taking each station's geographic coordinates as its position, producing a geographic view of the proposed network alongside the cluster maps from the selection stage.

4. Results

The majority of Hyperloop research focuses on implementation in Western Europe and North America. As such, there is unexplored potential in applications across three specific regions: West Africa, Southeast Asia, and the Gulf. These regions were chosen because they have the largest infrastructure gaps and the fastest-growing demand. Each corridor presents unique problems that Hyperloop can remediate – markets with no high-speed infrastructure at all, markets where existing infrastructure is already overwhelmed, and markets where freight demand is the primary reason for investment.

4.1. West Africa: Lagos-Accra Corridor (~550 km)

4.1.1. Socio Economic Context

Nigeria and Ghana are two of West Africa’s largest economies, with GDPs of around 252 billion USD (World Bank Group) and 82 billion USD, respectively [15], [16]. The countries’ combined urban population is expected to reach 173 million by 2050 [17]. Transport

contributes 5-8% of West Africa's GDP, yet the infrastructure serving it is severely underdeveloped [18]. Air passenger traffic across Africa is projected to grow from 160 million passengers per year in 2024 to nearly 500 million by 2050 [19]. Only 10-15% of African trade currently occurs between African countries, compared to roughly 60% within Europe, a gap that is, in part, attributable to poor transport connectivity.

4.1.2. Transportation Landscape

There is currently no passenger rail between Lagos and Accra. The corridor solely supports road travel, resulting in chronic delays at three international border crossings due to customs delays, requisite paperwork, inconsistent product standards, weak logistics coordination, and payment delays [20]. To commute by road takes between nine and fourteen hours. The largest current infrastructure project on this corridor is the Abidjan-Lagos Highway, a 1028 km road costing 15.6 billion USD. The highway is expected to begin construction in 2027 and be completed by 2033. The initiative is expected to generate over 30 thousand permanent jobs during operations [21]. Even with the addition of new road infrastructure, the necessity for scaled-up transport infrastructure is vital to revitalising West African commerce.

4.1.3. Regulatory and Policy Context

The Economic Community of West African States (ECOWAS) provides the existing treaty framework for transnational infrastructure in the region and has identified transport connectivity as a core priority in its Vision 2050 strategy [22]. Although no Hyperloop-specific regulations exist in the region, frameworks can be designed *ab initio* to fit such operations rather than adapting rail or aviation policy. The Abidjan-Lagos Highway project has already laid diplomatic groundwork across five nations and facilitates the opening of countries such as Burkina Faso, Mali and Niger [21]. Any agreements relating to Hyperloop would build on this foundation and need to address vehicle certification, safety standards, and revenue sharing between member states.

4.1.4. Benefits of Implementing Hyperloop

Akin to when West Africa skipped fixed telephone lines and went directly to mobile, the Lagos-Accra corridor could skip decades of incremental rail investment and move directly to high-speed infrastructure. The African Continental Free Trade Area (AfCFTA) is a single market of 1.3 billion people with a combined GDP of 3 trillion USD and depends on fast, reliable transport to function [23]. The planed highway alone is projected to cut transport costs by 48% and grow intra-regional trade by 15-25% [18]. A Hyperloop system has the potential to scale-up those same gains. The primary use case is freight: pharmaceuticals, agricultural goods, and manufactured products moving between Lagos and Tema Port near Accra.

4.2. Southeast Asia: Jakarta-Surabaya Corridor (~740 km)

4.2.1. Socio Economic Context

Java Island makes up just 7.25% of Indonesia's land mass but produces 57.9% of the country's GDP and is home to 159 million people. Jakarta's metropolitan economy has a GDP of 202.66 billion USD [24]. Surabaya is Indonesia's second-largest port city and industrial centre of East Java, with this corridor connecting the country's largest financial centre, main industrial zones, and busiest ports along a single axis, making it the most economically important domestic corridor in the country.

4.2.2. Transportation Landscape

The current train between Jakarta and Surabaya takes nine hours. in October 2023, Indonesia opened the Jakarta-Bandung High-Speed Railway, which was Southeast Asia's first high-speed line, cutting travel time between those two cities from over three hours to 46 minutes. The government is now assessing an extension of hat line to Surabaya, a route spanning over 700 km that would require major investment in land, tunnelling, and stations [25]. However, the existing Jakarta-Bandung line cost 7.3 billion USD and recorded losses of 258 million USD in 2024 alone [26].

4.2.3. Regulatory and Policy Context

Indonesia has no Hyperloop regulation, but the Ministry of Transportation's active work on the Surabaya HSR extension shows there is political will for large-scale rail investment [27]. A Hyperloop project would need new safety certification standards and financing structure that avoids the profitability problems visible in the Jakarta-Bandung project. China is the current technology and financing partner for Indonesia's HSR; a Hyperloop deployment would either need Chinese involvement or an alternative financing route through institutions like the Asian Deployment Bank (ADB). Indonesia's planned capital relocation to Nusantara in East Kalimantan also strengthens the long-term case for a broader Hyperloop network, rather than a single corridor, as economic activity shifts over the coming decades [28].

4.2.4. Benefits of Implementing Hyperloop

Jakarata-Surabaya serves as a strong example for Hyperloop as a demand-relief solution. The corridor's infrastructure is not reaching profitability, and the conventional high-speed rail model being proposed as the fix is showing severe financial strain before the extension has even been built. Hyperloop's advantage here is not speed alone.

4.3. The Middle East: The Gulf

4.3.1. Socio Economic Context

The Gulf Cooperation Council (GCC) is a six-country bloc with a combined GDP of approximately 2.3 trillion USD [29]. The Dubai-Riyadh corridor connects the two most economically active cities in the region. Dubai's GDP was approximately 120.6 billion USD in 2024 with a growth rate of 3.2% [30]. Over 95% of Dubai's GDP is non-oil based, with its transport and storage sector growing by 7.8% in Q2 2024, driven by logistics as a core pillar of the city's D33 economic strategy [31]. Saudi Arabia's GDP is approximately 1.24 trillion USD, with Vision 2030 committing hundreds of billions of USD to infrastructure and economic diversification [32]. The Dubai-Riyadh air route handled approximately 141 million passengers in 2025, rising 9.6% YoY [33]. This reflects strong demand and a complete absence of any surface transport alternative.

4.3.2. Transportation Landscape

There is no passenger rail between Dubai and Riyadh. The road journey takes approximately 12 hours, leaving air as the only practical option. The GCC Railway is a cross-border freight rail network linking all six member states and is around 50% complete with an operation target of 2030 [34]. It is a conventional freight rail system, not a high-speed solution. The ports of Jabel Ali, King Abdulaziz, and Salalah already position the Gulf as a major multi-modal trade hub for Asia-Europe-Africa flows [35]. Etihad Rail's Abu Dhabi-Dubai high-speed passenger service, launched in January 2025, is projected to contribute about 39 billion USD in GDP over 50 years [36].

5. Conclusion and Next Steps

The deliberation of corridor selection and passenger-freight allocation in this paper yielded an adaptive model that sufficiently addressed such questions as where a station could go, the feasibility and advantageousness, and overall if the specified location is a usable network. During deliberation, the stages were kept separate so that each could be revisited or refined, including swapping in different elevation data or altering the proximity threshold, as needed.

OpenStreetMap and Open-Elevation were chosen as the primary data sources. This decision was made based as both offer free, global, and reasonably current coverage of populated places, transport infrastructure, ports, and terrain; this is an important factor for a tool that is explicitly intended to be pointed at differing regions. The bounding box is the only input that requires change to move from one corridor to another. This directly supports the multi-region case study that was outlined for the Midwest corridor and Lagos–Accra or Jakarta–Surabaya corridors. The four enrichment attributes were chosen because each corresponds to a factor identified as being relevant to corridor and station viability: 1) Elevation variance reflects construction cost and difficulty, since flatter terrain is cheaper to build through, 2) Transport connectivity captures whether a location already has good access to air and rail, so that the network prioritizes locations where Hyperloop fills a genuine gap rather than duplicating existing service, 3) Port proximity reflects freight

integration potential, tying directly into the freight value proposition, and 4) Distance to the nearest major city acts as a demand proxy, reflecting the passenger time-compression value proposition.

The normalization of these attributes onto a common 0-1 scale before scoring was necessary in order to account for the metres of elevation spread, a count of nearby airports and stations, and the distance to the nearest port and city in kilometres. This method of scaling ensures that the scores would not be dominated by those possessing a larger numeric range, regardless of its intended importance. Inverting the normalized score for the three attributes where a lower raw value is better (elevation variance, port distance, and city distance) ensures that "higher score is always better" holds consistently across all four attributes, which simplifies both the scoring logic and the selection step that follows it.

The weight-profile structure, with separate weight sets for passenger, freight, and mixed operation, exists so that the same enriched dataset can be re-scored to reflect different strategic priorities without repeating any data collection. This is a direct implementation of the trade-offs between passenger and freight, where profiles and their weights can now be easily added and changed based on the needs of the Hyperloop network. K-means clustering was used for station selection to avoid common failure mode of naive top-scoring selection; taking the highest-scoring locations tends to produce a shortlist clustered around the same high-population area, since geographically close locations often score similarly. By first clustering candidates geographically and then selecting the highest-scoring location within each cluster, the tool guarantees a station layout that is spread across the corridor, closer to what an actual Hyperloop network would look like, while still preferring the best location available in each area.

Finally, representing the selected stations as a graph, with edges defined by a proximity threshold and weighted by geodesic distance, and computing routes with A* search using a geodesic heuristic, reflects that a Hyperloop network is fundamentally a graph problem once stations are fixed. Questions about which stations should be directly connected and what the shortest route between any two stations looks like are naturally expressed in graph

terms, and the geodesic-based weighting gives a simple, realistic baseline for both, directly supporting the network-level analysis.

While the current implementation selects station locations and connects them into a network based on proximity, it does not yet optimize the connections themselves. At present, any two selected stations within the proximity threshold are linked by an edge, with that edge weighted purely by geodesic distance. In practice, this means the network is close to fully connected, with every station linking to nearly every other station within range. Future iterations of this adaptive model should move beyond this and optimize which connections are built and implemented rather than only consider where stations currently sit. This is particularly important where elevation varies sharply between two otherwise close stations, since a direct route across mountainous terrain may be far more costly to construct than a longer route around it. The elevation variance attribute currently captures this only qualitatively, at the level of an individual station's surrounding terrain. A more complete model would incorporate elevation profiles along each candidate edge, allowing the system to favour or penalize specific connections based on the terrain they would need to cross, rather than treating all connections within the threshold as equivalent.

A second area for future development is real-time scheduling across the network. The current implementation produces a static graph and can compute a shortest path between two stations, but it does not model how passenger and freight vehicles would realistically be dispatched, sequenced, or prioritized over time. Given that Hyperloop capacity is managed through scheduling frequency rather than additional infrastructure, a future version of the tool should simulate real-time scheduling of mixed passenger and freight traffic across the network, including how that traffic interacts with, and potentially integrates with, other transport services such as high-speed rail, buses, and trucks at hub locations. This would extend the network from a static map of feasible connections into a dynamic model of how the system would operate day to day.

A third area is ticket pricing. The current tool focuses entirely on where infrastructure should be built and how it should be connected, with no model of how that infrastructure would be priced or operated commercially. A future version should incorporate a ticket pricing

system. This could be considered and enacted by comparing short-haul airfare and testing directly against specific stations and routes the tool selects, rather than relying on corridor-level estimates alone.

Finally, the attribute collection underlying station scoring should be made more rigorous. The current set of the four attributes portrays a useful but limited picture and is weighted toward broad economic and connectivity indicators. Future work should expand this to include a wider range of data, with elevation in particular collected at higher resolution and along candidate routes rather than only at station locations, alongside other terrain, environmental, and demographic attributes that were not within scope for the current implementation. A richer attribute set would allow the scoring and selection process to reflect a more complete picture of each location's suitability, rather than relying on a small number of proxy indicators.

References

[1] S. Premsagar and J. Kenworthy, “A Critical Review of Hyperloop (Ultra-High Speed Rail) Technology: Urban and Transport Planning, Technical, Environmental, Economic, and Human Considerations,” *Front. Sustain. Cities*, vol. 4, p. 842245, May 2022, doi: 10.3389/frsc.2022.842245.

[2] C. L. Taylor, D. J. Hyde, and L. C. Barr, “Hyperloop Commercial Feasibility Analysis: High Level Overview,” *Glenn Res. Cent. NASA*, no. DOT-VNTSC-NASA-16-01, Jul. 2016, [Online]. Available: https://rosap.ntl.bts.gov/view/dot/12308

[3] Directorate General for Mobility and Transport, Ramboll Management Consulting, TIS, CERTH, and SINTEF, “Fact-finding study on options for the possible further support of the European hyperloop sector: Final report.,” European Commission, Luxembourg, 2025. doi: 10.2832/3938374.

[4] Mid-Ohio Regional Planning Commission, “Economic benefits assessed in a feasibility study of a Midwest hyperloop system from Chicago to Pittsburgh total approximately $300 billion, with nearly $19 billion of that in direct transportation benefits.,” ITS Deployment Evaluation. Accessed: Jun. 13, 2026. [Online]. Available: https://www.itskrs.its.dot.gov/2020-b01462

[5] C. L. Taylor, D. J. Hyde, and L. C. Barr, “Hyperloop Commercial Feasibility Analysis: High Level Overview,” *Glenn Res. Cent. NASA*, no. DOT-VNTSC-NASA-16-01, Jul. 2016, [Online]. Available: https://rosap.ntl.bts.gov/view/dot/12308

[6] Transport Canada, “Preliminary Feasibility of Hyperloop Technology.” Accessed: Jun. 13, 2026. [Online]. Available: https://tc.canada.ca/en/innovation-centre/priority-reports/preliminary-feasibility-hyperloop-technology

[7] “A regulatory framework for hyperloop operation,” Zeleros. Accessed: Jun. 13, 2026. [Online]. Available: https://zeleros.com/regulatory-framework-for-hyperloop-operation/

[8] J. W. Visser, “Hyperloop as part of sustainable transport in Europe: Input for 2024 European Parliament Elections.” Hyperloop Development Program, Aug. 22, 2023. [Online]. Available: https://www.europarl.europa.eu/cmsdata/277661/HDP%20position%20paper%20European%20Parliament%20elections%202024.pdf

[9] Trinity Group, “GCC Unified Visa 2026: What It Means for Business Leaders.” Accessed: Jun. 13, 2026. [Online]. Available: https://trinitycorporateservices.com/gcc-unified-visa-2026/

[10] J. Varghese, “Unified GCC Visa now coming in 2026: What’s behind the new launch date,” Gulf News. Accessed: Jun. 13, 2026. [Online]. Available: https://gulfnews.com/business/tourism/unified-gcc-visa-now-coming-in-2026-whats-behind-the-new-launch-date-1.500339495

[11] A. Vesjolijs, Y. Stukalina, and O. Zervina, “A Framework for Quantifying Hyperloop’s Socio-Economic Impact in Smart Cities Using GDP Modeling,” *Economies*, vol. 13, no. 8, p. 228, Aug. 2025, doi: https://doi.org/10.3390/economies13080228.

[12] Z. Grujic and B. Grujic, “Optimal Routing in Urban Road Networks: A Graph-Based Approach Using Dijkstra’s Algorithm,” *Appl. Sci.*, vol. 15, no. 8, p. 4162, Apr. 2025, doi: 10.3390/app15084162.

[13] G. A. V. Pai, *Mastering Advanced Data Structures and Algorithm Design Strategies*, 1st ed. in A Textbook of Data Structures and Algorithms, no. 3. Wiley, 2023. doi: 10.1002/9781394192014.

[14] H. Ortega-Arranz, D. R. Llanos, and A. Gonzalez-Escribano, *The Shortest-Path Problem: Analysis and Comparison of Methods*. in Synthesis Lectures on Theoretical Computer Science. Cham: Springer International Publishing, 2015. doi: 10.1007/978-3-031-02574-7.

[15] “GDP (current US$) - Nigeria,” World Bank Open Data. Accessed: Jun. 13, 2026. [Online]. Available: https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=NG

[16] “GDP (current US$) - Ghana,” World Bank Open Data. Accessed: Jun. 13, 2026. [Online]. Available: https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=GH

[17] African Development Bank, “African Development Bank and partners plan to make Abidjan-Lagos corridor highway a potent economic and industrial hub,” African Development Bank Group. Accessed: Jun. 13, 2026. [Online]. Available: https://www.afdb.org/en/news-and-events/press-releases/african-development-bank-and-partners-plan-make-abidjan-lagos-corridor-highway-potent-economic-and-industrial-hub-78940

[18] AIF, “US$15bn Lagos-Abidjan highway project receives lifeline,” Africa Investment Forum. Accessed: Jun. 13, 2026. [Online]. Available: https://www.africainvestmentforum.com/en/news/news/us15bn-lagos-abidjan-highway-project-receives-lifeline-432

[19] N. Ekugo, “Lagos-Abidjan transport corridor requires urgent investment to close $30bn infrastructure gap- Official,” Businessday NG. Accessed: Jun. 13, 2026. [Online]. Available: https://businessday.ng/news/article/lagos-abidjan-transport-corridor-requires-urgent-investment-to-close-30bn-infrastructure-gap-official/

[20] E. C. Macaulay, “Africa signed a trade deal. The real border problem never left,” Businessday NG. Accessed: Jun. 13, 2026. [Online]. Available: https://businessday.ng/opinion/article/africa-signed-a-trade-deal-the-real-border-problem-never-left/

[21] ECOWAS Commission, “About the Project,” Abidjan-Lagos Corridor Highway Development Project. Accessed: Jun. 13, 2026. [Online]. Available: https://alchdp.com/about/

[22] A. S. Adekanbi, “Critical Infrastructure Development in West Africa: A Strategic Imperative for Regional Integration and Economic Growth,” *Afr. Policy Res. Inst.*, Nov. 2025, doi: https://doi.org/10.59184/pb025.11.

[23] AIF, “The Africa Investment Forum: catalyzing financing for game-changing Abidjan-Lagos highway project.” Accessed: Jun. 13, 2026. [Online]. Available: https://www.africainvestmentforum.com/en/news/news/africa-investment-forum-catalyzing-financing-game-changing-abidjan-lagos-highway-project

[24] BPS-Statistics Indonesia, “Jakarta’s Economic Growth in The Fourth Quarter of 2024,” BPS-Statistics Indonesia. Accessed: Jun. 13, 2026. [Online]. Available: https://jakarta.bps.go.id/en/pressrelease/2025/02/05/1198/jakartaseconomic-growth-in-the-fourth-quarter-of-2024.html

[25] P. B. Santoso, “Indonesia Pushes Ahead with 780-Km Jakarta–Surabaya High-Speed Rail Plan,” Jakarta Globe. Accessed: Jun. 13, 2026. [Online]. Available: https://jakartaglobe.id/news/indonesia-pushes-ahead-with-780km-jakartasurabaya-highspeed-rail-plan

[26] S. Strangio, “Indonesia High-speed Rail Project a Financial ‘Time Bomb,’ Official Says,” *The Diplomat*, Aug. 21, 2025. Accessed: Jun. 13, 2026. [Online]. Available: https://thediplomat.com/2025/08/indonesia-high-speed-rail-project-a-financial-time-bomb-official-says/

[27] B. Dey, “Indonesia Eyes Nation-Shaping High-Speed Rail Link From Jakarta To Surabaya As Part Of A Long-Term Vision For Sustainable Economic Growth, Urban Integration, And Reduced Congestion Across Java - Travel And Tour World.” Accessed: Jun. 13, 2026. [Online]. Available: https://www.travelandtourworld.com/news/article/indonesia-eyes-nation-shaping-high-speed-rail-link-from-jakarta-to-surabaya-as-part-of-a-long-term-vision-for-sustainable-economic-growth-urban-integration-and-reduced-congestion-across-java/

[28] S. McEwan and A. Skinner, “A capital is born: The impact of Indonesia moving its capital city,” Oxford Economics. Accessed: Jun. 13, 2026. [Online]. Available: https://www.oxfordeconomics.com/resource/a-capital-is-born-the-impact-of-indonesia-moving-its-capital-city/

[29] Qatar News Agency, “GCC GDP Ranked Ninth Globally, Exceeds $2 Trillion in 2024, GCC-Stat Data Show,” Qatar News Agency. Accessed: Jun. 13, 2026. [Online].

Available: https://qna.org.qa/en/news/news-details?id=gcc-gdp-ranked-ninth-globally-exceeds-2-trillion-in-2024-gcc-stat-data-show&date=1%2F02%2F2026

[30] “Dubai Economy - dda - AE,” dda. Accessed: Jun. 13, 2026. [Online]. Available: https://www.dubai.ae/dubai-economy

[31] G. N. Report, “Dubai’s Q2 2024 GDP rises 3.3% to reach Dh116 billion,” Gulf News: Latest UAE news, Dubai news, Business, travel news, Dubai Gold rate, prayer time, cinema. Accessed: Jun. 13, 2026. [Online]. Available: https://gulfnews.com/business/markets/dubais-q2-2024-gdp-rises-33-to-reach-dh116-billion-1.1731232924525

[32] “GDP (current US$) - Saudi Arabia,” World Bank Open Data. Accessed: Jun. 13, 2026. [Online]. Available: https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=SA

[33] M. Lane, “Saudi Arabia hits record air passenger numbers in 2025,” Moodie Davitt Report. Accessed: Jun. 13, 2026. [Online]. Available: https://moodiedavittreport.com/saudi-arabia-hits-record-air-passenger-numbers-in-2025/

[34] J. Mairura, “A Closer Look into Middle East’s Longest GCC Railway Project and Completion Timeline,” Construction Review. Accessed: Jun. 13, 2026. [Online]. Available: https://constructionreviewonline.com/construction-of-middle-easts-longest-gcc-railway-project-on-track-for-2030-completion/

[35] “UAE Transportation Infrastructure Construction Market - Size, Share & Trends,” Mordor Intelligence. Accessed: Jun. 13, 2026. [Online]. Available: https://www.mordorintelligence.com/industry-reports/uae-transportation-infrastructure-construction-market

[36] “GCC Freight and Logistics Market Size & Growth to 2031,” Mordor Intelligence. Accessed: Jun. 13, 2026. [Online]. Available: https://www.mordorintelligence.com/industry-reports/gcc-freight-and-logistics-market

Appendix 1: Node Scoring Weight Profile

Passenger Profile:

Attribute	**Weight**
Elevation Variance	0.15
Transport Access Count	0.25
Distance to Port (km)	0.4
Distance to Major City (km)	0.2

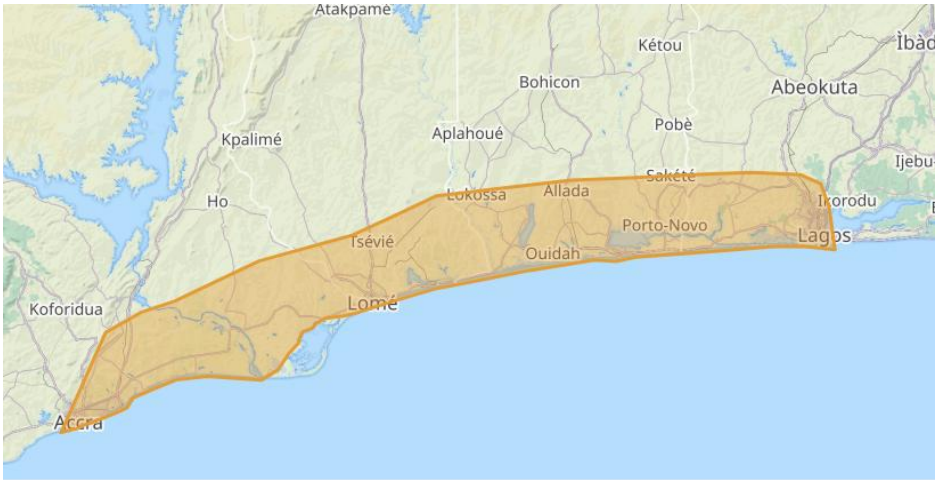
Freight Profile:

Attribute	**Weight**
Elevation Variance	0.15
Transport Access Count	0.20
Distance to Port (km)	0.15
Distance to Major City (km)	0.50

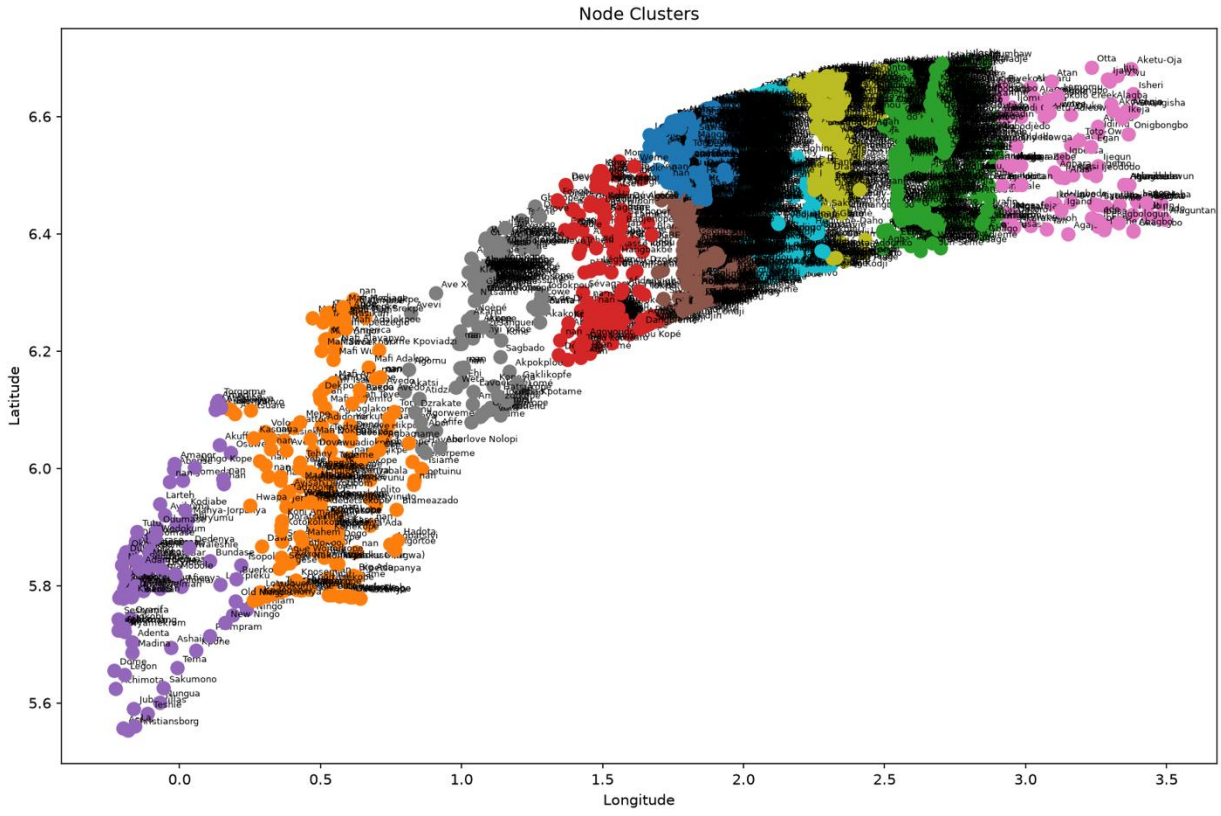
Mixed Profile:

Attribute	**Weight**
Elevation Variance	0.2
Transport Access Count	0.25
Distance to Port (km)	0.25
Distance to Major City (km)	0.3

Appendix 2: Lagos-Accra corridor geographic region of interest



Appendix 3: Lagos-Accra Corridor: Node Clusters Output



Appendix 4: Lagos–Accra Corridor - Optimal Hyperloop Node Network

